



YOUR KINDLE NOTES FOR:

## The Three-Body Problem (Remembrance of Earth's Past)

by Cixin Liu, Ken Liu

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### 117 Highlights | 3 Notes

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Highlight (Yellow) | Page 15

“Lin Lin is too smart. To work in fundamental theory, one must be stupid.” In later years, I began to understand his words more and more. Lin, you truly are too smart. Even a few years ago, you could feel the political winds shifting in academia and prepared yourself. For example, when you taught, you changed the names of many physical laws and constants: Ohm’s law you called resistance law, Maxwell’s equations you called electromagnetic equations, Planck’s constant you called the quantum constant.... You explained to your students that all scientific accomplishments resulted from the wisdom of the working masses, and those capitalist academic authorities only stole these fruits and put their names on them.

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Highlight (Yellow) | Page 16

“In China, any idea that dared to take flight would only crash back to the ground. The gravity of reality is too strong.”

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Highlight (Yellow) | Page 18

“You: you’re trying to say that God exists?” “I don’t know.” “What?” “I’m saying I don’t know. If by ‘God’ you mean some kind of superconsciousness outside the universe, I don’t know if it exists or not. Science has given no evidence either way.”

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Highlight (Yellow) | Page 24

Her company wielded hundreds of chain saws like a swarm of steel locusts, and after they passed, only stumps were left.

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Highlight (Yellow) | Page 27

Both the ocean and the iceberg are made of the same material. That the iceberg seems separate is only because it is in a different form. In reality, it is but a part of the vast ocean....

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Highlight (Yellow) | Page 28

It was impossible to expect a moral awakening from humankind itself, just like it was impossible to expect humans to lift off the earth by pulling up on their own hair. To achieve moral awakening required a force outside the human race.

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Highlight (Yellow) | Page 38

“Wenjie, let me put all my cards on the table. Your case has a lot of prosecutorial discretion. On the one hand, we could minimize it as a case of an educated youth being fooled by a reactionary book—it’s not a big deal. We don’t even need to go through a judicial procedure. We’ll have you attend a political class and write a few self-criticism reports, and then you can go back to the Construction Corps. On the other hand, we could also prosecute this case to its fullest extent. Wenjie, you must know that you could be declared an active counter-revolutionary. “Now, faced with political cases like yours, all prosecutorial organs and courts would rather be too severe than too lax. This is because treating you too severely would just be a mistake in method, but treating you too laxly would be a mistake in political direction. Ultimately, however, the decision belongs to the military control commission. Of course, I’m telling you all this off the record.”

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Highlight (Yellow) | Page 43

Yang said that he wanted to focus on experimental and applied problems, staying away from theory. Ye Wenjie recalled her father saying, “I’m not opposed to your idea. But we are, after all, the department of theoretical physics. Why do you want to avoid theory?” Yang replied, “I want to devote myself to the times, to make some real-world contributions.” Her father said, “Theory is the foundation of application. Isn’t discovering fundamental laws the biggest contribution to our time?” Yang hesitated and finally revealed his real concern: “It’s easy to make ideological mistakes in theory.” Her father had nothing to say to that.

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Highlight (Yellow) | Page 43

Yang was very talented, with a good mathematical foundation and a quick mind. But during his brief time as a graduate student, he always kept a respectable distance from his thesis advisor.

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Highlight (Yellow) | Page 62

“Do you think we’ll forget you have a mouth if you don’t use it all the time?”

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Highlight (Yellow) | Page 62

Shi’s questions irritated Wang again, not unlike the feeling a man has upon finding out that he has swallowed a fly with his meal.

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Highlight (Yellow) | Page 64

a person’s ability to discern the truth is directly proportional to his knowledge.”

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Highlight (Yellow) | Page 66

have you ever had anything happen to you that changed your life completely? Some event where afterward the world became a totally different place for you?”

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Highlight (Yellow) | Page 69

“We ran the same experiment five times. Four of the experiments differed in both location and time. Two of the experiments were at the same location but different times. Aren’t you shocked by the results?” He opened his arms exaggeratedly. “Five times! Every colliding experiment yielded the exact same result!”

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Highlight (Yellow) | Page 69

“Can you explain this incredible result? Please use the language of physics.” “All right ... During these five experiments, the mass of the two balls never changed. In terms of their locations, as long as we’re using the frame of reference of the tabletop, there was also no change. The velocity of the white ball striking the black ball also remained basically the same throughout. Thus, the transfer of momentum between the two balls didn’t change. Therefore, in all five experiments, the result was the black ball being driven into the pocket.”

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Highlight (Yellow) | Page 69

We’ve discovered a great principle of nature: The laws of physics are invariant across space and time. All the physical laws of human history, from Archimedes’ principle to string theory, and all the scientific discoveries and intellectual fruits of our species are the by-products of this great law. Compared to us two theoreticians, Einstein and Hawking are mere applied engineers.”

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Highlight (Yellow) | Page 70

“Imagine another set of results. The first time, the white ball drove the black ball into the pocket. The second time, the black ball bounced away. The third time, the black ball flew onto the ceiling. The fourth time, the black ball shot around the room like a frightened sparrow, finally taking refuge in your jacket pocket. The fifth time, the black ball flew away at nearly the speed of light, breaking the edge of the pool table, shooting through the wall, and leaving the Earth and the Solar System, just like Asimov once described.<sup>13</sup> What would you think then?”

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Highlight (Yellow) | Page 72

You really believe that the laws of physics are not invariant across time and space?” “I don’t know anything.” Ding stared into Wang’s eyes for a long time. Finally, he said, “But that is the question.”

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Highlight (Yellow) | Page 74

two words “shooter” and “farmer.” This was a reference to two hypotheses, both involving the fundamental nature of the laws of the universe. In the shooter hypothesis, a good marksman shoots at a target, creating a hole

every ten centimeters. Now suppose the surface of the target is inhabited by intelligent, two-dimensional creatures. Their scientists, after observing the universe, discover a great law: “There exists a hole in the universe every ten centimeters.” They have mistaken the result of the marksman’s momentary whim for an unalterable law of the universe. The farmer hypothesis, on the other hand, has the flavor of a horror story: Every morning on a turkey farm, the farmer comes to feed the turkeys. A scientist turkey, having observed this pattern to hold without change for almost a year, makes the following discovery: “Every morning at eleven, food arrives.” On the morning of Thanksgiving, the scientist announces this law to the other turkeys. But that morning at eleven, food doesn’t arrive; instead, the farmer comes and kills the entire flock. Wang felt the road beneath his feet shift like quicksand. The A-shaped building seemed to wobble and sway. He quickly brought his gaze back to the street.

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Highlight (Yellow) | Page 82

He glanced at the monitor. The game site’s URL caught his attention: [www.3body.net](http://www.3body.net).

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Highlight (Yellow) | Page 90

finally, the main screen reflected the main reaction chamber’s halted status. Almost simultaneously, the countdown before Wang’s eyes also stopped. The final number was 1174:10:07. A few seconds later, the numbers flickered and disappeared.

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Highlight (Yellow) | Page 112

the makers of Three Body took the exact opposite of the approach taken by designers of other games. Normally, game designers tried to display as much information as possible to increase the sense of realism. But Three Body’s designers worked to compress the information content to disguise a more complex reality, just like that seemingly empty photograph of the sky.

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Highlight (Yellow) | Page 116

“But she was a woman. A woman should be like water, able to flow over and around anything.”

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Highlight (Yellow) | Page 120

I know that as the universe cooled after the big bang, the leftover ‘embers’ became the cosmic microwave background. The radiation fills the entire universe and can be observed in the centimeter wavelength range. I think it was back in the sixties when two Americans accidentally discovered the radiation when they were testing a supersensitive satellite reception antenna—”

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Highlight (Yellow) | Page 127

The red light had come from more than ten billion years ago. It was the remnants of the big bang, the still-warm embers of Creation. He could not see any stars. Normally, since visible light would be compressed to invisible by the glasses, each star should appear as a black dot. But the diffraction of centimeter-wave radiation overwhelmed all other shapes and details.

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Highlight (Yellow) | Page 132

I'm a simple man without a lot of complicated twists and turns. Look down my throat and you can see out my ass.

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Highlight (Yellow) | Page 133

"But, I did indeed invent an ultimate rule." "Tell me." "Anything sufficiently weird must be fishy." "What ... what kind of crappy rule is that?" "I'm saying that there's always someone behind things that don't seem to have an explanation."

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Highlight (Yellow) | Page 135

"Everything that's happening is coordinated by someone behind the scenes with one goal: to completely ruin scientific research." "Who?" "I have no idea. But I can sense the plan, a very comprehensive, intricate plan: damage scientific research installations, kill scientists, drive scientists like you crazy and make you commit suicide—but the main goal is to misdirect your thoughts until you're even more foolish than ordinary people." "Your last statement is really perceptive."

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Highlight (Yellow) | Page 137

I know General Chang very well. He's the sort who's afraid of nothing, not even the sky falling, but I can tell that he's worried about something much worse right now. They're all scared out of their wits, and they have no confidence that we'll win." "If what you say is true, then we should all be frightened." "Everyone is afraid of something. The enemy must be, too. The more powerful they are, the more they have to lose to their fears." "What do you think the enemy is afraid of?" "You! Scientists! The odd thing is that the less practical your research is, the more they're afraid of you—like abstract theories, the kind of thing Yang Dong worked on. They are more frightened of such work than you are of the universe winking at you. That's why they're so ruthless."

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Highlight (Yellow) | Page 138

Keep your head screwed on straight, and if you get scared again, just remember my ultimate rule."

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Highlight (Yellow) | Page 150

By the time you're my age, you'll realize that everything you once thought mattered so much turns out to mean very little."

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Highlight (Yellow) | Page 150

In his current state, his mental stability depended on two pillars: this old woman, who had weathered so many storms and become as gentle as water, and Shi Qiang, the man who feared nothing because he knew nothing.

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Highlight (Yellow) | Page 150

“Many of those retirees were my colleagues back at the university, but somehow I just can’t mix with them. Everyone likes to reminisce, but no one wants to listen, and everyone feels annoyed when someone else tells a story.

Highlight (Yellow) | Page 154

if the supervisor said to go east, they would work hard to move west, purposely playing the fool. Their hope was to put the following thought into the supervisor’s head: This man is working hard, but he’s limited in his skills. There’s no point in keeping him, because he’ll only get in the way. Many really did successfully obtain transfers through this method.

Highlight (Yellow) | Page 156

“Do you know about microwave ovens?” Ye shook her head. “They are a luxury plaything of the capitalist West. Food is heated by the energy generated from absorbing microwave radiation. At my previous research station, in order to precisely test the high-temperature aging of certain components, we imported one. After work, we would use it to warm mantou bread, bake a potato, that sort of thing. It’s very interesting: The inside heats up first while the outside remains cold.” Commissar Lei stood up and paced back and forth. He was so close to the edge of the cliff that it made Ye nervous. “Red Coast is a microwave oven, and its heating targets are the enemy’s space vessels. If we can apply microwave radiation at a specific power level of one-tenth of a watt to one watt per square centimeter, we’ll be able to disable or destroy many electronic components of satellite communications, radar, and navigation systems.”

Highlight (Yellow) | Page 157

“The emission from the system seems to be modulated.” “That’s right. However, the modulation is unlike that used in conventional radio communications. The purpose isn’t to add information, but to use shifting frequencies and amplitudes to penetrate possible shielding by the enemy.

Highlight (Yellow) | Page 157

“Recently, two target satellites were launched from Jiuquan. The test attacks by Red Coast were completely successful. Temperature inside the satellites reached nearly a thousand degrees, and all instruments and photographic equipment onboard were destroyed. In future wars, Red Coast can effectively strike at the enemy’s communication and reconnaissance satellites, like the KH-8 spy satellites on which the American Imperialists rely, as well as the KH-9, which are about to be launched. The lower-orbit spy satellites of the Soviet Revisionists are even more vulnerable. If necessary, we even have the capacity to destroy the Salyut space station of the Soviet Revisionists and the Skylab station that American Imperialists plan on launching next year.”

Highlight (Yellow) | Page 166

One or more technological leaps can occur at any time. Such an occurrence will pose a major threat to our strategic planning.

Highlight (Yellow) | Page 166

Fields where technological leaps are most likely: Physics: [omitted] Biology: [omitted] Computer Science: [omitted] The Search for Extraterrestrial Intelligence (SETI): Of all fields, this is the one in which the possibility for a technology leap is greatest. If a leap occurs in this field, the impact will exceed the sum of technology leaps in the other three fields.

Highlight (Yellow) | Page 170

Using the elemental code above and supplemented with low-resolution images, gradually build up to a full linguistic system. Languages supported: Chinese and Esperanto. The entire system's information content should be 680 KB. Transmission times at the 2,800 MHz, 12,000 MHz, and 22,000 MHz channels are 1,183 minutes, 224 minutes, and 132 minutes respectively.

Highlight (Yellow) | Page 170

Content of Message Transmitted by Red Coast [Summary] Overview of Earth (3.1 KB), overview of life on Earth (4.4 KB), overview of human society (4.6 KB), basic world history (5.4 KB). Total information content: 17.5 KB. The entire message will be sent after transmitting the self-interpreting code system. Transmission times of message at the 2,800 MHz, 12,000 MHz, and 22,000 MHz channels are 31 minutes, 7.5 minutes, and 3.5 minutes, respectively. The message will be carefully vetted by a multidisciplinary review to ensure that it will not give away the Earth's coordinates relative to the Milky Way. Among the three channels, transmission at the higher-frequency 12,000 MHz and 22,000 MHz channels should be minimized to reduce the likelihood that the source of transmission may be precisely ascertained.

Highlight (Yellow) | Page 171

Fourth Draft [Complete Text] We extend our best wishes to you, inhabitants of another world. After reading the following message, you should have a basic understanding of civilization on Earth. By dint of long toil and creativity, the human race has built a splendid civilization, blossoming with a multitude of diverse cultures. We have also begun to understand the laws governing the natural world and the development of human societies. We cherish all that we have accomplished. But our world is still flawed. Hate exists, as does prejudice and war. Because of conflicts between the forces of production and the relations of production, wealth distribution is extremely uneven, and large portions of humanity live in poverty and misery. Human societies are working hard to resolve the difficulties and problems they face, striving to create a better future for Earth civilization. The country that sent this message is engaged in this effort. We are dedicated to building an ideal society, where the labor and value of every member of the human race are fully respected, where everyone's material and spiritual needs are fully met, so that civilization on Earth may become more perfect. With the best of intentions, we look forward to establishing contact with other civilized societies in the universe. We look forward to working together with you to build a better life in this vast universe.

Highlight (Yellow) | Page 175

Naïve, idealistic hopes had been shattered. Scholars found that, contrary to the happy wishes of most people, it was not a good idea for the human race as a whole to make contact with extraterrestrials. The impact of such

contact on human society would be divisive rather than uniting, and would exacerbate rather than mitigate the conflicts between different cultures. In summary, if contact were to occur, the internal divisions within Earth civilization would be magnified and likely lead to disaster. The most shocking conclusion of all was that the impact would have nothing at all to do with the degree and type of contact (unidirectional or bidirectional), or the form and degree of advancement of the alien civilization.

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Highlight (Yellow) | Page 180

“From time to time, I would gaze up at the stars after a night shift and think that they looked like a glowing desert, and I myself was a poor child abandoned in the desert.... I thought that life was truly an accident among accidents in the universe. The universe was an empty palace, and humankind the only ant in the entire palace. This kind of thinking infused the second half of my life with a conflicted mentality: Sometimes I thought life was precious, and everything was so important; but other times I thought humans were insignificant, and nothing was worthwhile. Anyway, my life passed day after day accompanied by this strange feeling, and before I knew it, I was old....”

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Highlight (Yellow) | Page 183

“I don’t have a calendar. I only brought a model built upon observation data. But I must make it clear that even if the model is correct, it’s not certain that by using it one can master the precise details of the sun’s motion and create a calendar. However, it’s a necessary step.”

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Highlight (Yellow) | Page 184

“It’s actually pretty simple. The reason why the sun’s motion seems patternless is because our world has three suns. Under the influence of their mutually perturbing gravitational attraction, their movements are unpredictable—the three-body problem. When our planet revolves around one of the suns in a stable orbit, that’s a Stable Era. When one or more of the other suns move within a certain distance, their gravitational pull will snatch the planet away from the sun it’s orbiting, causing it to wander unstably through the gravitational fields of the three suns. That’s a Chaotic Era. After an uncertain amount of time, our planet is once again pulled into a temporary orbit and another Stable Era begins. This is a football game at the scale of the universe. The players are the three suns, and our planet is the football.”

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Highlight (Yellow) | Page 188

in this civilization, Copernicus successfully revealed the basic structure of the universe. The civilization of Three Body will take its first leap. The game has now entered the second level. We invite you to log on to the second level of Three Body.

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Highlight (Yellow) | Page 190

I knew that I had some special talents others lacked. For example, if you drew a line, I could always draw another line that would divide it into the golden ratio: 1.618.

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Highlight (Yellow) | Page 190

The Collected Cases of Sherlock Holmes. He turned to one story— “A Study in Scarlet,” I think. There’s a scene in it where Watson sees a plainly dressed messenger downstairs and points him out to Holmes. Holmes says, “Oh, you mean the retired sergeant of marines?” Watson is amazed by how Holmes could deduce the man’s history, but Holmes can’t articulate his reasoning and has to think for a while to figure out his chain of deductions. It was based on the man’s hand, his movements, and so on. He tells Watson that there is nothing strange about this: Most people would have difficulty explaining how they know two and two make four. My teacher closed the book and said to me, “You’re just like that. Your derivation is so fast and instinctive that you can’t even tell how you got the answer.”

Highlight (Yellow) | Page 193

So I created a sphere in this infinite space for myself: not too big, though possessing mass. My mental state didn’t improve, however. The sphere floated in the middle of “emptiness”—in infinite space, anywhere could be the middle. The universe had nothing that could act on it, and it could act on nothing. It hung there, never moving, never changing, like a perfect interpretation for death. I created a second sphere whose mass was equal to the first one’s. Both had perfectly reflective surfaces. They reflected each other’s images, displaying the only existence in the universe other than itself. But the situation didn’t improve much. If the spheres had no initial movement—that is, if I didn’t push them at first—they would be quickly pulled together by their own gravitational attraction. Then the two spheres would stay together and hang there without moving, a symbol for death. If they did have initial movement and didn’t collide, then they would revolve around each other under the influence of gravity. No matter what the initial conditions, the revolutions would eventually stabilize and become unchanging: the dance of death. I then introduced a third sphere, and to my astonishment, the situation changed completely. Like I said, any geometric figure turns into numbers in the depths of my mind. The sphereless, one-sphere, and two-sphere universes all showed up as a single equation or a few equations, like a few lonesome leaves in late fall. But this third sphere gave “emptiness” life. The three spheres, given initial movements, went through complex, seemingly never-repeating movements. The descriptive equations rained down in a thunderstorm without end. Just like that, I fell asleep.

Highlight (Yellow) | Page 194

The physics principles behind the three-body problem<sup>28</sup> are very simple. It’s mainly a math problem.

Highlight (Yellow) | Page 209

“Have you heard the joke about the Army of Sichuan?” Von Neumann took out a thick cigar, bit off the end, and lit it. “Some soldiers were being drilled, but because they had no education, they couldn’t even follow the drill instructor’s simple orders to march LEFT-RIGHT-LEFT. So the instructor came up with a solution: He had every soldier wear a straw shoe on the left foot and a cloth shoe on the right. When they marched, he shouted”—here he switched to a Sichuan accent—“STRAW-CLOTH-STRAW-CLOTH.... That’s the kind of soldier we need. Except we need thirty million of them.”

Highlight (Yellow) | Page 211

even the complicated objects of the universe are made from the simplest elements. I only need three.”

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Highlight (Yellow) and Note | Page 225

The location for the Three Body players’ meet-up was a small, out-of-the-way coffee shop. Wang had always imagined game meet-ups would be lively events full of people, but this meet-up consisted of only seven players, including himself. Like Wang, the other six did not look like gaming enthusiasts.

### **Not credible**

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Highlight (Yellow) | Page 228

the bodies of the Trisolarans who formed the computer were covered by a purely reflective surface, which probably evolved as a response to survival under extreme conditions of sunlight. The mirrorlike surface could be deformed into any shape, and they communicated with each other by focusing light with their bodies. This kind of light-speech could transmit information extremely rapidly and was the foundation of the Trisolaran-formation computer.

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Highlight (Yellow) | Page 233

The bit that Newton lacked was the effect on planetary orbit from the gravitationally induced curvature of space-time described by general relativity. Though the error caused by it was small, its impact on the results of the computation was fatal. Adding the correction factor for perturbation from space-time curvature to the classical equations would yield the right mathematical model. The amount of computational power required far exceeds what you accomplished in the East, but is easily provided by modern computers.”

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Highlight (Yellow) | Page 237

A few days later, one of the suns moved to a distance where its outer gaseous layer became visible. In the middle of a tranquil night, the star suddenly turned into a blazing sun. Separated by intervals of thirty hours or so, the other two suns also appeared in quick succession. This was not a normal kind of tri-solar day. By the time the last flying star turned into a sun, the first sun had already swept past the planet at extremely close range. Right after that, the other two suns swept past Trisolaris at even closer ranges, well within the planet’s Roche limit<sup>31</sup>, such that the tidal forces imposed on Trisolaris by the three suns exceeded the force of the planet’s gravitational self-attraction. The first sun shook the deepest geological structure of the planet; the second sun tore open a great rift in the planet that went straight to the core; and the third sun ripped the planet into two pieces.”

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Highlight (Yellow) | Page 238

After the planet was torn apart, the two irregularly shaped pieces each returned to spherical form under self-gravitation.

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Highlight (Yellow) | Page 238

and as the geology of the mother planet settled down, evolution began its tottering steps in new oceans and on new continents, until civilization reappeared for the one hundred and ninety-second time. The entire process took ninety million years.

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Highlight (Yellow) | Page 240

Does it represent the yearning for order, or the surrender to chaos?

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Highlight (Yellow) and Note | Page 242

Civilization 192 was a milestone in Trisolaran civilization. It finally proved that the three-body problem had no solution. It gave up the useless effort that had already lasted through 191 cycles and set the course for future civilizations. Thus, the goal of Three Body has changed. The new goal is: Head for the stars; find a new home.

**Even if calculatable... The later conclusion to head for the strs seems prudent. Then question is why not us even if exponentially more stable though perhaps answer is the relative risk of said travel**

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Highlight (Yellow) | Page 244

In most regions of the Milky Way, the distribution of stars is fairly even. It's the result of star clusters acting under the influence of gravity. The distance between most stars is between three and six light-years."

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Highlight (Yellow) | Page 262

The sun was an amplifier for radio waves.

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Highlight (Yellow) | Page 262

in order to reach the energy mirrors, the waves would have to be more powerful than a threshold value. The vast majority of Earth-based radio sources could not cross this threshold, but the Jovian radio outburst did—

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Highlight (Yellow) | Page 263

The problem with solar outages was not resolved, but another exciting possibility presented itself: Humans could use the sun as a superantenna, and, through it, broadcast radio waves to the universe. The radio waves would be sent with the power of the sun, hundreds of millions of times greater than the total usable transmission power on Earth. Earth civilization had a way to transmit at the level of a Kardashev Type II civilization.

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Highlight (Yellow) | Page 270

As an astrophysicist, Ye was strongly against nuclear weapons. She knew this was a power that should belong only to the stars. She knew also that the universe had even more terrible forces: black holes, antimatter, and

more. Compared to those forces, a thermonuclear bomb was nothing but a tiny candle. If humans obtained mastery over one of those other forces, the world might be vaporized in a moment.

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Highlight (Yellow) | Page 276

The message that was winging its way to the sun said, Come here! I will help you conquer this world. Our civilization is no longer capable of solving its own problems. We need your force to intervene.

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Highlight (Yellow) | Page 277

“This ... Lord that you talk about. Why is it so afraid of nanomaterial?” “Because it can allow humans to escape gravity and engage in space construction at a much larger scale.” “The space elevator?” Wang suddenly understood. “Yes. If ultrastrong nanomaterials could be mass produced, then that would lay the technical foundation for building a space elevator from the ground up to a geostationary point in space. For our Lord, this is but a tiny invention; but for humans on Earth, its meaning would be significant. With this technology, humans could easily enter near-Earth space and build up large-scale defensive structures. Thus, this technology must be extinguished.”

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Highlight (Yellow) | Page 293

They all treated Ye with great respect and were very polite toward her. She didn't pay much attention to this at first. But after a while, after observing how those men roughly beat their wives and flirted outrageously with the widows in the village, saying things that made her blush, she finally realized how precious their respect was. Every few days, one of them would bring a hare or pheasant he had caught to Hunter Qi's home. They also gave Yang Dong strange and quaint toys that they'd made with their own hands. In Ye's memory, these months seemed to belong to someone else, like a segment of another life that had drifted into hers like a feather. This period condensed in her memory into a series of classical paintings—not Chinese brush paintings but European oil paintings. Chinese brush paintings are full of blank spaces, but life in Qijiatus had no blank spaces. Like classical oil paintings, it was filled with thick, rich, solid colors. Everything was warm and intense: the heated kang stove-beds lined with thick layers of ura sedge, the Guandong and Mohe tobacco stuffed in copper pipes, the thick and heavy sorghum meal, the sixty-five-proof baijiu distilled from sorghum—all of these blended into a quiet and peaceful life, like the creek at the edge of the village.

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Highlight (Yellow) | Page 297

It was such an important thing, and yet she had done it all by herself. This gave her a sense of unreality. As time passed, that sense grew ever stronger. What had happened resembled an illusion, a dream. Could the sun really amplify radio signals? Did she really use it as an antenna to send a message about human civilization into the universe? Did she really receive a message from the stars? Did that blood-hued morning, when she had betrayed the entire human race, really happen? And those murders ...

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Highlight (Yellow) | Page 306

“That afternoon, I went to see the hellish coast. The sea was black, and the waves, under the sticky, thick film of oil, were smooth and weak. The beach was also covered by a black layer of crude oil. Some volunteers and I

searched for birds on the beach that were still alive. They struggled in the sticky oil, looking like black statues made out of asphalt, only their eyes proving that they were still alive. Those eyes staring out of the oil still haunt my dreams to this day. We soaked those birds in detergent, trying to get rid of the oil stuck to their bodies. But it was extremely difficult: crude oil was infused into their feathers, and if you brushed a little too hard, the feathers would come off with the oil.... By that evening, most of the birds had died. As I sat on the black beach, exhausted and covered in oil, I stared at the sun setting over a black sea and felt like it was the end of the world.

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Highlight (Yellow) | Page 306

“My father then said, ‘Mike, I’ve told you how dinosaurs went extinct. An asteroid crashed into the Earth. The world first became a sea of fire, and then sank into a prolonged period of darkness and coldness.... One night, you woke from a nightmare, saying that you had dreamt that you were back in that terrifying age. Let me tell you now what I wanted to tell you that night: If you really lived during the Cretaceous Period, you’d be fortunate. The period we live in now is far more frightening. Right now, species on Earth are going extinct far faster than during the late Cretaceous. Now is truly the age of mass extinctions! So, my child, what you’re seeing is nothing. This is only an insignificant episode in a much vaster process. We can have no sea birds, but we can’t be without oil. Can you imagine life without oil? Your last birthday, I gave you that lovely Ferrari and promised you that you could drive it after you turned fifteen. But without oil, it would be a pile of junk metal and you’d never drive it. Right now, if you want to visit your grandfather, you can get there on my personal jet and cross the ocean in a dozen hours or so. But without oil, you’d have to tumble in a sailboat for more than a month.... These are the rules of the game of civilization: The first priority is to guarantee the existence of the human race and their comfortable life. Everything else is secondary.’

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Highlight (Yellow) | Page 307

“My father placed a great deal of hope in me, but in the end I didn’t turn out the way he wanted. In the days after that, the eyes of those drowned birds always followed me and determined my life. When I was thirteen, my father asked me what I wanted to do when I grew up. I said I wanted to save lives. My dream wasn’t that great. I only wanted to save a species near extinction.

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Highlight (Yellow) | Page 307

‘All lives are equal’ is the basic tenet of Pan-Species Communism.” “What?” Ye wasn’t sure she had heard the last term correctly. “Pan-Species Communism. It’s an ideology I invented. Or maybe you can call it a faith. Its core belief is that all species on Earth are created equal.” “That is an impractical ideal. Our crops are also living species. If humans are to survive, that kind of equality is impossible.” “Slave owners must also have thought that about their slaves in the distant past. And don’t forget technology—there will be a day when humanity can manufacture food. We should lay down the ideological and theoretical foundation long before that. Indeed, Pan-Species Communism is a natural continuation of the Universal Declaration of Human Rights. The French Revolution was two hundred years ago, and we haven’t even taken a step beyond that. From this we can see the hypocrisy and selfishness of the human race.”

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Highlight (Yellow) | Page 308

On the way back, one of Ye's colleagues recited from Chairman Mao's essay "Remembering Bethune": "Noble-minded and pure, a man of moral integrity and above vulgar interests." He sighed. "There really are people who can live like that." Others also expressed their admiration and conflicted feelings. Ye seemed to be speaking to herself as she said, "If there were more men like him, even just a few more, things would have turned out differently." Of course, no one understood what she really meant.

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Highlight (Yellow) | Page 308

"Why not? Of the four possible sites, this has the best electromagnetic environment." "What about the human environment? Comrades, don't just focus on the technical side. Look at how poor this place is. The poorer a village, the craftier the people. Do you understand? If the observatory were located here, there would be trouble between the scientists and the locals. I can imagine the peasants thinking of the astronomy complex as a juicy piece of meat that they can take bites from." This site was indeed not approved, and the reason was just what the task force leader had said.

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Highlight (Yellow) | Page 309

"That's true. The county actually gave him an award for planting the trees. The villagers originally planned to cut down the forest in a few more years—it's best to wait until the pig is fat before slaughtering it, am I right? But those people from Nange Village can't wait any longer, and if my village doesn't join in, we won't get any." "You must stop immediately. I will go to the government to report this!" "There's no need." The village chief lit a cigarette and pointed to a truck loading the cut trees in the distance. "See that? That's from the deputy secretary of the County Forestry Bureau. And there are also people here from the town police department. They've carried off more trees than anyone else!"

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Highlight (Yellow) | Page 310

Not including fixed assets, do you know how much money I have at my disposal? About four point five billion dollars. I could easily ask them to stop and get them to plant more trees. I could make all the loess hills within sight be covered by quick-growth forest. But what would be the point? "Everything you see before you is the result of poverty. But how are things any better in the wealthy countries? They protect their own environments, but then shift the heavily polluting industries to the poorer nations. You probably know that the American government just refused to sign the Kyoto Protocol.... The entire human race is the same. As long as civilization continues to develop, the swallows I want to save and all the other swallows will go extinct. It's just a matter of time."

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Highlight (Yellow) | Page 311

"Do you know why I came here?" Evans continued. "The seeds of Pan-Species Communism had sprouted long ago in the ancient East." "You're thinking of Buddhism?" "Yes. The focus of Christianity is Man. Even though all the species were placed into Noah's Ark, other species were never given the same status as humans. But Buddhism is focused on saving all life. That was why I came to the East. But ... it's obvious now that

everywhere is the same.” “Yes, that’s true. Everywhere, people are the same.” “What can I do now? What is the purpose of my life? I have four point five billion dollars and an international oil company. But what good is all that? Humans have surely invested more than forty-five billion dollars in saving species near extinction. And probably more than four hundred and fifty billion has already been spent on saving the environment from degradation. But what’s the use? Civilization continues to follow its path of destruction of all life on Earth except humans. Four point five billion is enough to build an aircraft carrier, but even if we build a thousand aircraft carriers, it would be impossible to stop the madness of humanity.”

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Highlight (Yellow) | Page 312

Evans said, “I still can’t believe what you just told me. It’s too fantastic. But luckily, I have the resources to confirm this. If what you told me is true”—he extended his hand and spoke the words that every new member of the future ETO would have to say upon joining—“let us be comrades.” 28 The Second Red Coast Base Three more years passed.

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Highlight (Yellow) | Page 314

“The great Trisolaran Fleet has already set sail. Their target is this solar system, and they will arrive in four hundred and fifty years.” Ye remained calm. Nothing could surprise her anymore. Evans pointed to the crowd behind him. “You’re looking at the first members of the Earth-Trisolaris Organization. Our ideal is to invite Trisolaran civilization to reform human civilization, to curb human madness and evil, so that the Earth can once again become a harmonious, prosperous, sinless world. More and more people identify with our ideal, and our organization is growing rapidly. We have members all over the world.” “What can I do?” Ye asked in a soft voice. “You will become the commander in chief of the Earth-Trisolaris Movement. This is the wish of all ETO fighters.”

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Highlight (Yellow) | Page 314

Evans raised a fist and shouted at the crowd, “Eliminate human tyranny!” Accompanied by the sound of crashing waves and the wind howling against the antenna, the ETO fighters shouted as one, “The world belongs to Trisolaris!” This was the day that the Earth-Trisolaris Movement formally began.

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Highlight (Yellow) | Page 320

most of them believed that the ideal solution would be to find a way to allow the Lord to continue to live in the Trisolaris stellar system and avoid the invasion of the Earth. Naïvely, they believed that solving the three-body problem would achieve this goal, saving both Trisolaris and the Earth. Admittedly, perhaps this thought wasn’t all that naïve. Trisolaran civilization itself had thought so through many eons. The effort to solve the three-body problem was a thread that ran through several hundreds of cycles of Trisolaran civilization. Most Redemptionists with some in-depth math and physics knowledge had attempted the three-body problem, and even after knowing that the problem was mathematically unsolvable as posed, the effort did not cease, because solving the three-body problem had become a religious ritual of their faith. Even though the Redemptionists had many first-class physicists and mathematicians, research in this area never yielded any important results. It took someone like Wei Cheng, a prodigy who had no connection to the ETO or the Trisolaran faith, to accidentally come up with a breakthrough in which the Redemptionists placed much hope.

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Highlight (Yellow) | Page 327

“Let’s focus on one thing first. The Trisolarans were able to shoot two protons at the Earth from four light-years away and they both reached the target! That accuracy is incredible! There are numerous obstacles between there and here: interstellar dust, for example. And both the solar system and the Earth are moving. It would require more precision than shooting a mosquito here from Pluto. The shooter is beyond imagination.”

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Highlight (Yellow) | Page 328

If you take out all the active charcoal, it will look like a little pile of black particles, like mouse droppings. But added together, the adsorbent surface formed by the tiny holes inside is as large as a tennis court. This is why active charcoal is so adsorbent.” “What are you trying to say?” Wang asked, listening intently. “The sponge or active charcoal inside a filter is three-dimensional. Their adsorbent surfaces, however, are two-dimensional. Thus, you can see how a tiny high-dimensional structure can contain a huge low-dimensional structure. But at the macroscopic level, this is about the limit of the ability for high-dimensional space to contain low-dimensional space. Because God was stingy, during the big bang He only provided the macroscopic world with three spatial dimensions, plus the dimension of time. But this doesn’t mean that higher dimensions don’t exist. Up to seven additional dimensions are locked within the micro scale, or, more precisely, within the quantum realm. And added to the four dimensions at the macro scale, fundamental particles exist within an eleven-dimensional space-time.”

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Highlight (Yellow) | Page 329

Making use of fundamental particles without taking advantage of the micro dimensions is something that our naked, hairy ancestors already began back when they lit bonfires within caves. Controlling chemical reactions is just manipulating micro particles without regard to the micro dimensions. Of course, this control also progressed from crude to advanced: from bonfires to steam engines, and then generators. Now, the ability for humans to manipulate micro particles at the macro level has reached a peak: We have computers and nanomaterials. But all of that is accomplished without unlocking the many micro dimensions. From the perspective of a more advanced civilization in the universe, bonfires and computers and nanomaterials are not fundamentally different. They all belong to the same level. That’s also why they still think of humans as mere bugs. Unfortunately, I think they’re right.” “Can you be more specific? What does all this have to do with those two protons? Ultimately, what can the two protons that have reached the Earth do?

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Highlight (Yellow) | Page 350

The only way to escape this fate was to mate with a member of the opposite sex. When that happened, the organic material making up their bodies would meld into one. Two-thirds of the material would then become fuel to power the biochemical reaction that would completely renew the cells in the remaining one-third and create a new body. Then this body would divide into three to five tiny new lives: their children. They would inherit some of the memories of their parents, continue their lives, and begin the cycle of life anew.

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Highlight (Yellow) and Note | Page 352

Listening Post 1379 had sent out a warning message in the direction of the transmission.

**If they were so intelligent... Listening stations wouldn't (unilaterally) transmit.**

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Highlight (Yellow) | Page 352

The mental states that Trisolarians needed were calmness and numbness. The history of the past two hundred-some cycles of civilization proved that civilizations that relied on these two states as their spiritual core were the most capable of survival.

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Highlight (Yellow) | Page 353

But there is another reason for destroying Earth civilization. They're also a warlike race. Very dangerous. If we try to coexist with them on the same planet, they will shortly learn our technology. Continuing in that state would allow neither civilization to thrive. Let me ask you: You wish to be the savior of the Earth, but do you not feel any sense of responsibility for your own race?"

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Highlight (Yellow) | Page 353

"The kind of civilization you yearn for once existed on Trisolaris, too. They had free, democratic societies, and they left behind rich cultural legacies. You know barely anything about them. Most details have been sealed away and forbidden from view. But in all the cycles of Trisolaran civilization, this type of civilization was the weakest and most short-lived. A modest Chaotic Era disaster was enough to extinguish them. Look again at the Earth civilization that you wish to save. A society born and bred in the eternal spring of a beautiful hothouse would not be able to survive even a million Trisolaran hours if it were transplanted here." "That flower may be delicate, but it possesses peerless splendor. She enjoys freedom and beauty in the ease of paradise."

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Highlight (Yellow) | Page 358

"We've all studied the first messages from Earth carefully. The section most worthy of attention is their history. Let's observe the facts: Humans took more than a hundred thousand Earth years to progress from the Hunter-Gatherer Age to the Agricultural Age. To get from the Agricultural Age to the Industrial Age took a few thousand Earth years. But to go from the Industrial Age to the Atomic Age took only two hundred Earth years. Thereafter, in only a few Earth decades, they entered the Information Age. This civilization possesses the terrifying ability to accelerate their progress. "On Trisolaris, of the more than two hundred civilizations, including our own, none has ever experienced such accelerating development. The progress of science and technology in all Trisolaran civilizations has been at a constant or decelerating pace. In our world, each technology age requires approximately the same amount of time for steady, slow development." The princeps nodded. "The fact is that four million and five hundred thousand hours from now, when the Trisolaran Fleet has reached the Earth, that civilization's technology level will have long surpassed ours, due to their accelerating development. The journey of the Trisolaran Fleet is long and arduous, and the fleet must pass through two interstellar dust belts. It's very likely that only half of the ships will reach the Earth's solar system, while the rest

perish along the way. And then, the Trisolaran Fleet will be at the mercy of a much more powerful Earth civilization. This is not an expedition, but a funeral procession!”

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Highlight (Yellow) | Page 359

The princeps said, “What we must do next is contain the progress of science on Earth. Luckily, as soon as we received the first messages from Earth, we began to develop plans to do so.

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Highlight (Yellow) | Page 360

“The plan focuses on emphasizing the negative environmental effects of scientific development and showing signs of supernatural power to the population of Earth. In addition to highlighting the negative effects of progress, we’ll also attempt to use a series of ‘miracles’ to construct an illusory universe that cannot be explained by the logic of science. After these illusions have been maintained for some time, it’s possible that Trisolaran civilization may become a target of religious worship there. Then, unscientific ways of thinking will dominate scientific thinking among human intellectuals, and lead to the collapse of the entire scientific system of thought.”

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Highlight (Yellow) | Page 361

Overall technological development depends on the advancement of basic science, and the foundation of basic science lies in the exploration of the deep structure of matter. If there’s no progress in this field, there can be no major breakthrough in science and technology as a whole. Of course, this is not specific to civilization on Earth. It is applicable to all targets

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Highlight (Yellow) | Page 361

“Project Sophon, to put it simply, aims to transform a proton into a superintelligent computer.”<sup>42</sup>

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Highlight (Yellow) | Page 362

The etching of micro integrated circuits can only occur at the macro scale, and only on a macroscopic two-dimensional plane. Thus, we must unfold a proton into two dimensions.” “Unfold a nine-dimensional structure into two dimensions? How big would the area be?” “Very big, as you will see.” The science consul smiled.

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Highlight (Yellow) | Page 365

“How complicated can the internal structure of a subatomic particle be?” “It depends on the number of dimensions of your observation perspective. From a one-dimensional perspective, it’s only a point—that’s how ordinary people think of the particles. From a two- or three-dimensional perspective, the particle begins to show internal structure. From a four-dimensional perspective, a fundamental particle is an immense world.” The princeps said, “To use a word like ‘immense’ to describe a subatomic particle such as a proton seems incredible to me.” The science consul ignored the princeps and continued, “As we move to higher dimensions, the complexity and number of structures within a particle increase dramatically. The comparisons I’m about to make

will not be precise, but should give you an idea of the scale. A particle seen from a seven-dimensional perspective has a complexity comparable to our Trisolaran stellar system in three dimensions. From an eight-dimensional perspective, a particle is a vast presence like the Milky Way. When the perspective has been raised to nine dimensions, a fundamental particle's internal structures and complexity are equal to the whole universe. As for even higher dimensions, our physicists haven't been able to explore them, so we cannot yet imagine the degree of complexity."

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Highlight (Yellow) | Page 366

"Do these show that the microcosmos contained within the unfolded proton harbors intelligent life?" "Our definition of 'life' is probably not appropriate for the high-dimensional microcosmos. More accurately, we can only say that universe contains intelligence or wisdom. Scientists have long predicted this possibility. It would have been odd for such a complex and vast world to not have evolved something akin to intelligence."

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Highlight (Yellow) | Page 368

"You have not mentioned the real danger. What if the proton is unfolded into zero dimensions?" "Zero dimensions?" The princeps was interested. "Wouldn't that be a point with no size?" "Yes, a singularity! Even a proton would be infinitely big compared to it. The entire mass of the proton will be contained in this singularity, and its density will be infinite. Princeps, I'm sure you can imagine what that would be." "A black hole?" "Yes." "Princeps,

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"Did we destroy a civilization in the microcosmos in this experiment?" "It was at least an intelligent body. Also, Princeps, we destroyed the entire microcosmos. That miniature universe is immense in higher dimensions, and it probably contained more than one intelligence or civilization that never had a chance to express themselves in macro space. Of course, in higher dimensional space at such micro scales, the form that intelligence or civilization may take is beyond our imagination. They're something else entirely. And such destruction has probably occurred many times before." "Oh?" "In the long history of scientific progress, how many protons have been smashed apart in accelerators by physicists? How many neutrons and electrons? Probably no fewer than a hundred million. Every collision was probably the end of the civilizations and intelligences in a microcosmos. In fact, even in nature, the destruction of universes must be happening at every second—for example, through the decay of neutrons. Also, a high-energy cosmic ray entering the atmosphere may destroy thousands of such miniature universes.... You're

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I will immediately notify the propaganda consul and direct him to repeatedly publicize this scientific fact to the world. The people of Trisolaris must understand that the destruction of civilizations is a common occurrence that happens every second of every hour." "Why? Do you wish to encourage the people to face the possible destruction of Trisolaran civilization with equanimity?" "No. It's to encourage them to face the destruction of Earth civilization with equanimity. You know very well that after we publicized our policy toward the Earth civilization, there was a wave of extremely dangerous pacifism. We have only now discovered that there are many like the listener of Post 1379. We must control and eliminate these weak sentiments."

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The principles governing micro-scale integrated circuits were completely different from those of conventional circuits, as the base material wasn't made of atoms, but matter from a single proton. The "p-n junctions" of the circuits were formed by twisting the strong nuclear forces locally on the surface of the proton plane, and the conducting lines were made of mesons that could transmit the nuclear force. Because the surface area for the circuit was extremely large, the circuits were also very large. The circuit lines were as thick as hairs, and an observer close enough could see them with the naked eye. Flying close to the proton membrane, it could be seen as a vast plane made of complex, elaborate integrated circuits. The total area covered by the circuits was dozens of times the area of the continents on Trisolaris.

Highlight (Yellow) | Page 374

Micro-Intelligence 2.10 loaded. Sophon One ready to accept commands. The science consul said, "A sophon has been born. We have endowed a proton with wisdom. This is the smallest artificial intelligence that we can make." "But right now, it appears as the largest artificial intelligence," said the princeps. "As soon as we increase the dimensionality of this proton, it will become very small."

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"Can you increase its dimensionality to eleven, and make it as small as a regular proton?" As soon as the princeps said this, the science consul shouted to the sophon, his voice tinged with fear, "Attention! This is not a command!" Sophon One understands. "Princeps, if we increased the dimensionality to eleven, we would lose it forever. When the sophon shrinks to the size of a regular subatomic particle, the internal sensors and I/O ports will be smaller than the wavelength of any electromagnetic radiation. That means it would not be able to sense the macro world, and would not be able to receive our commands." "But we must eventually make it shrink back to a subatomic particle." "Yes, but that must await the completion of Sophon Two, Sophon Three, and Sophon Four. Multiple sophons may be able to form a system to sense the macro world through quantum effects. For example, suppose a nucleus has two protons. The two of them will interact and follow certain patterns of motion. Take spin: Maybe the direction of spin of the two protons must be opposite from each other. When these two protons are taken out of the nucleus, no matter how far apart they are, this pattern will remain in effect. When both protons are made into sophons, they will, based on this effect, create a mutual-sensing system. More sophons can then form a mutual-sensing formation. This formation's scale can be adjusted to any size, and can thus receive electromagnetic waves to sense the macro world at any frequency. Of course, the actual quantum effects necessary to create such a sophon formation are very complicated. My explanation is only an analogy."

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>Sophon formation, adjust dimensionality to eleven. After the command was issued, the four mirrored spheres disappeared. The science consul said, "Princeps, now Sophon One and Sophon Two will be launched toward the Earth. Using the large knowledge base stored in the micro circuits, the sophons understand the nature of space. They can draw energy from the vacuum and become high-energy particles in a moment, and navigate through space at nearly the speed of light. This might appear to violate the law of conservation of energy, but in fact the sophons are only 'borrowing' energy from the structure of vacuum. However, the time for returning such energy is far in the future, when the proton decays. By then, the end of the universe will not be far.

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In actual experiments, they use the substance containing the target particles as the bull's-eye for the accelerated bullets. "But the inside of the substance being struck is almost all vacuum. Suppose an atom is the size of a theater; the nucleus is like a walnut hovering in the center of the theater. Thus, successful collisions are very rare. Often large quantities of high-energy particles must be directed against the target substance for a sustained period of time before a collision occurs. This kind of experiment is akin to looking for a raindrop of a slightly different color in a summer thunderstorm. "This gives the sophons an opening. A sophon can take the place of a target particle and accept the collision. Because they're highly intelligent, they can precisely determine through the quantum sensing formation the paths that the accelerated particles will follow within a very short period of time and move to the appropriate location. Thus, the likelihood that a sophon will be struck will be billions of times greater than the actual target particle. After a sophon is struck, it can deliberately give out wrong and chaotic results. Thus, even if the actual target particle is occasionally struck, Earth physicists will not be able to tell the correct result from the numerous erroneous results." "Wouldn't this destroy the sophon as well?" asked the military consul. "No. When a sophon is smashed into several pieces, several new sophons are born. And they continue to have secure quantum entanglements between them, just like how, if you break a magnet in half, you would get two magnets. Even though each partial sophon's capabilities will be much lower than the original, whole sophon, under the direction of the self-healing software, the pieces will move together and reassemble into the original sophon. This process only requires a microsecond and will occur after the collision in the accelerator, and after the pieces of the sophon have left the wrong results in the bubble chamber or on sensitive film."

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Someone asked, "Would it be possible for Earth scientists to find a way to detect sophons and then use a strong magnetic field to imprison them? Protons have positive charge." "That's impossible. To detect sophons requires humans to make breakthroughs in the study of the deep structure of matter. But their high-energy accelerators will all have been turned into heaps of junk. How can they make progress in such research? The hunter's eyes have already been blinded by the prey he intends to catch."

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Indeed, just these two are more than enough, because each sophon is capable of multitasking." "Multitasking?" "This is a bit of jargon related to ancient serial computers. Back then, a computer's central processing unit could only carry out a single instruction at a time. But, because it was so fast, and aided by interrupt scheduling, from our low-speed perspective, the computer was carrying out multiple programs at the same time. As you know, the sophons move at close to the speed of light. The surface of the Earth is a tiny space for sophons. If sophons patrol around the accelerators on Earth at this speed, then, from the perspective of humans, it is as if they simultaneously exist in all the accelerators and can almost simultaneously create erroneous results in all the accelerators. "By our calculations, each sophon is capable of controlling more than ten thousand high-energy accelerators. It takes about four to five years for humans to build each of these accelerators,

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“Sophons can create miracles?” “For humans, yes. Everyone knows that high-energy particles can expose film. This is one of the ways that primitive accelerators on Earth once showed individual particles. When a sophon passes through the film at high energy, it leaves behind a tiny exposed spot. If a sophon passes back and forth through the film many times, it can connect the dots to form letters or numbers or even pictures, like embroidery. The process is very fast, and far quicker than the speed at which humans expose film when taking a picture. Also, the human retina is similar to the Trisolaran one. Thus, a high-energy sophon can also use the same technique to show letters, numbers, or images on their retina.... And if these little miracles can confuse and terrify humans, then the next great miracle will be sufficient to frighten their scientists—no better than bugs—to death: Sophons can cause background cosmic radiation to flash in their eyes.” “This would be very frightening for our scientists as well. How would this be accomplished?” “Very simple. We have already written the software to allow a sophon to unfold itself into two dimensions. After the unfolding is complete, the huge plane can wrap itself around the Earth. This software can also adjust the membrane so that it’s transparent, but the degree of transparency can be tuned in the frequencies of the cosmic microwave background.... Of course, as sophons fold and unfold into different dimensions, they can display even more amazing ‘miracles.’ The software for accomplishing these is still being developed, but these ‘miracles’ will create a mood sufficient to divert human scientific thought onto the wrong path. This way, we can use the Miracle Plan to effectively restrain scientific endeavors outside of physics on Earth.”

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Highlight (Yellow) | Page 382

Why not send all four of the completed sophons to Earth?” “Quantum entanglement can work at a distance. Even if four sophons were placed at opposite ends of the universe, they could still sense each other instantaneously, and the quantum formation between them would still exist. Keeping Sophon Three and Sophon Four here will enable them to receive the information sent back by Sophon One and Sophon Two instantaneously. This gives us a way to monitor the Earth in real time. Also, the sophon formation allows Trisolaris to communicate in real time with the alienated forces within Earth civilization.”

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Highlight (Yellow) | Page 382

“Comrades, please be aware that our meeting is probably already being monitored by sophons. From now on, there will be no more secrets.” When he said this, the surroundings were still familiar. The shadows of summer trees swayed against the drawn curtains, but in the eyes of the attendees, the world was no longer the same. They felt the gaze of omnipresent eyes. Under these eyes, there was nowhere to hide in the world. This feeling would follow them all their lives, and their descendants would not be able to escape it. It would take many, many years before humans finally made the mental adjustment to this situation. Three seconds after General Chang finished his remark, Trisolaris communicated with humanity outside the ETO for the first time. After this, they terminated all communications with the Adventists. For the remainder of the lives of all attendees, Trisolaris never sent another message.

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Highlight (Yellow) | Page 384

Everyone in the Battle Command Center saw the message in their eyes, just like Wang Miao’s countdown. The message flashed into existence for only two seconds and then disappeared, but everyone got it. It was only a single sentence: You’re bugs!

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Highlight (Yellow) | Page 387

“I just want to ask the two of you one question: Is the technological gap between humans and Trisolarans greater than the one between locusts and humans?” The question hit the two scientists like a bucket of cold water. As they stared at the clumps of locusts before them, their expressions grew solemn. They got Shi Qiang’s point.

\* \* \* Look at them, the bugs. Humans have used everything in their power to extinguish them: every kind of poison, aerial sprays, introducing and cultivating their natural predators, searching for and destroying their eggs, using genetic modification to sterilize them, burning with fire, drowning with water. Every family has bug spray, every desk has a flyswatter under it ... this long war has been going on for the entire history of human civilization. But the outcome is still in doubt. The bugs have not been eliminated. They still proudly live between the heavens and the earth, and their numbers have not diminished from the time before the appearance of the humans. The Trisolarans who deemed the humans bugs seemed to have forgotten one fact: The bugs have never been truly defeated.

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Highlight (Yellow) | Page 393

I’ve always felt that the greatest and most beautiful stories in the history of humanity were not sung by wandering bards or written by playwrights and novelists, but told by science. The stories of science are far more magnificent, grand, involved, profound, thrilling, strange, terrifying, mysterious, and even emotional, compared to the stories told by literature. Only, these wonderful stories are locked in cold equations that most do not know how to read. The creation myths of the various peoples and religions of the world pale when compared to the glory of the big bang. The three-billion-year history of life’s evolution from self-reproducing molecules to civilization contains twists and romances that cannot be matched by any myth or epic. There is also the poetic vision of space and time in relativity, the weird subatomic world of quantum mechanics ... these wondrous stories of science all possess an irresistible attraction. Through the medium of science fiction, I seek only to create my own worlds using the power of imagination, and to make known the poetry of Nature in those worlds, to tell the romantic legends that have unfolded between Man and Universe.

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Highlight (Yellow) | Page 394

The wonder of science fiction is that it can, when given certain hypothetical world settings, turn what in our reality is evil and dark into what is righteous and bright, and vice versa. This book and its two sequels try to do just that, but no matter how reality is twisted by imagination, it ultimately remains there.

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perhaps tomorrow we’ll wake up and find an alien spaceship the size of the moon parked in orbit. The appearance of extraterrestrial intelligence will force humanity to confront an Other. Before then, humanity as a whole will never have had an external counterpart. The appearance of this Other, or mere knowledge of its existence, will impact our civilization in unpredictable ways.

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